Two-fragment pairing model for sub-barrier neutron pair transfer reactions between superfluid nuclei

Neutron-pair transfer between superfluid nuclei selectively probes nuclear pairing and the coherent dynamics of correlated nucleons. Quantum simulation can retain this coherence and particle-number conservation, but established reaction models map poorly onto near-term hardware. We introduce a number-conserving two-fragment pairing model in which a closest-approach-dependent coupling transfers correlated pairs between finite pairing spaces. Its quasi-spin representation yields a compact qubit Hamiltonian. With fitted coupling parameters, exact model-space evolution reproduces transfer data for ^{116}Sn + ^{60}Ni $\rightarrow$ ^{114}Sn + ^{62}Ni and ^{96}Zr + ^{40}Ca $\rightarrow$ ^{94}Zr + ^{42}Ca and quantifies interfragment entanglement. This work establishes a classically verifiable benchmark and provides a controlled foundation for more realistic quantum simulations of sub-barrier nuclear reactions.

I. INTRODUCTION

Pairing correlations are an essential feature of atomic nuclei that strongly influence their many-body structure and dynamics. In open-shell nuclei, the attractive interaction between like nucleons near the Fermi surface favors the formation of coherent pairs in time-reversed single-particle states. These pairs predominantly couple to a total angular momentum $J = 0$, giving rise to a finite-nucleus analogue of Bardeen-Cooper-Schrieffer superfluidity. The critical importance of this phenomenon is evident through widespread experimental signatures across the nuclear chart, shaping binding energies through odd-even mass staggering, altering decay properties, and producing low-lying spectra characterized by collective pairing excitations and reduced rotational moments of inertia. Furthermore, understanding these microscopic correlations is essential for astrophysics, as their bulk counterpart governs the macroscopic superfluid response of neutron-star matter.

Neutron pair transfer reactions provide a highly selective experimental probe of these correlations. In these reactions, the transfer of a neutron pair intimately connects neighboring nuclei. Due to pairing near the Fermi surface, the transfer process preferentially populates specific final states [4]. Consequently, comparing measured cross sections with reaction models yields a quantitative measure of the pairing correlations [4]. Measurements performed at sub-barrier energies provide a particularly clean reaction regime, as the spatial separation of the colliding nuclei effectively suppresses fusion and complex dissipative processes [5, 6]. The transfer thus takes place between nuclei moving along nearly pure Coulomb trajectories with large separations, making it exclusively sensitive to the extreme periphery of the nucleus. This straightforward reaction mechanism provides a direct link between the experimental cross sections and the surface pairing properties [5, 6].

While several theoretical frameworks successfully model neutron pair transfer [4, 10–12], their central approximations often obscure the underlying coherent many-body dynamics. Channel models describe only the initial and final states, while mean-field approaches often fail to conserve particle number and limit quantum entanglement. Quantum simulation provides a unitary, number-conserving alternative that explicitly captures these crucial correlations and neutron-transfer distributions [8, 9]. A direct implementation of established reaction models is nevertheless impractical on present quantum hardware because their distorted continuum waves, optical potentials, three-dimensional spatial grids, large quasiparticle spaces, and symmetry projections do not furnish a compact finite qubit Hamiltonian. These considerations motivate the two-fragment pairing model introduced here, which isolates the coherent pair dynamics into a compact, number-conserving quasi-spin form that maps directly onto qubits.

The scope of the present study is deliberately limited. The present model remains small enough to be solved exactly on a classical computer and therefore neither demonstrates nor claims quantum advantage. Instead, it constitutes a controlled first step toward quantum simulations of more realistic transfer reactions, which will require larger single-particle spaces, broken-pair configurations, and additional reaction channels. Its immediate advantage is that it can be implemented on near-term quantum hardware: particle-number conservation and the quasi-spin representation map the time-dependent collision Hamiltonian onto a finite qubit Hamiltonian composed of single-qubit terms and two-qubit couplings [8, 9]. Classical tractability permits stringent validation of the quantum implementation, while the hardware-compatible construction provides a benchmark for progressively more complete descriptions of nuclear reactions.

The paper is organized as follows. Section II formulates the two-fragment pairing model and its time-dependent collision dynamics. Section III presents numerical calculations for ^{116}Sn+^{60}Ni$\rightarrow$ ^{114}Sn + ^{62}Ni, evaluates the calculated pair-transfer probabilities against experiment, quantifies the interfragment entanglement, and explores the model's application to ^{96}Zr+^{40}Ca$\rightarrow$ ^{94}Zr + ^{42}Ca. Section IV summarizes the findings and outlines future steps toward more realistic quantum simulations of nuclear reactions.

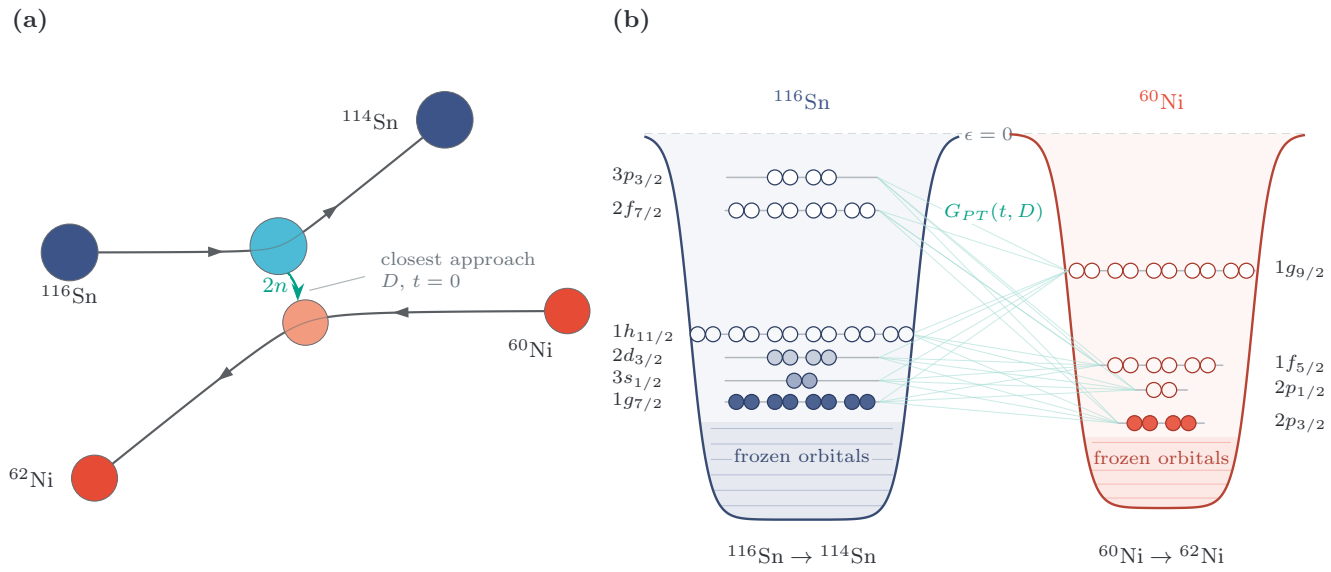


FIG. 1. Two-fragment representation of sub-barrier neutron-pair transfer. (a) Semiclassical Rutherford trajectories in the center-of-mass frame for $^{116}\mathrm{Sn} + {}^{60}\mathrm{Ni} \to {}^{114}\mathrm{Sn} + {}^{62}\mathrm{Ni}$, with a correlated neutron pair exchanged near the distance of closest approach D. (b) Finite seniority-zero spaces embedded in schematic neutron Woods–Saxon mean fields. The labeled horizontal levels are the active orbitals listed in Table I. Each closely spaced circle doublet represents one neutron-pair slot, and its fill denotes the exact entrance-ground-state occupation $\langle \hat{n}_{pX} \rangle / \Omega_{pX}$. The pale crossing network denotes the uniform Hermitian $J = 0$ pair-transfer coupling $G_{PT}(t, D)$ between every active projectile and target orbital. The potential profiles and horizontal level extents are illustrative. The shaded lower regions and their horizontal levels denote frozen occupied orbitals below the active spaces.

II. TWO-FRAGMENT PAIRING MODEL

A. Pairing description of the fragments

We represent the projectile ($X = P$) and target ($X = T$) by finite sets of active neutron orbitals, as summarized in Fig. 1. An orbital p has single-particle angular momentum j_{pX} and pair degeneracy $\Omega_{pX} = (2j_{pX}+1)/2$. The operator that creates a $J = 0$ pair in time-reversed states is

$$\hat{P}^{\dagger}_{pX} = \sum_{m>0} (-1)^{j_{pX}-m} \hat{a}^{\dagger}_{pmX} \hat{a}^{\dagger}_{p,-m,X}, \tag{1}$$

where $\hat{a}^{\dagger}_{pmX}$ creates a neutron with magnetic projection m in orbital p of fragment X; the restriction $m > 0$ counts each time-reversed doublet once. We restrict the dynamics to seniority zero, so every active neutron belongs to such a pair. A fragment basis state is then specified by its orbital pair occupations,

$$\begin{aligned} &|\boldsymbol{n}_X\rangle = |n_{1X}, \ldots, n_{L_X,X}\rangle\,, \\ &0 \le n_{pX} \le \Omega_{pX}, \qquad M_X = \sum_{p\in X} n_{pX}. \end{aligned} \tag{2}$$

where L_X and M_X denote, respectively, the number of active orbitals and the total number of active pairs in fragment X. The quasi-spin algebra gives

$$\begin{aligned} \hat{P}^{\dagger}_{pX} |n_{pX}\rangle &= \sqrt{(\Omega_{pX} - n_{pX})(n_{pX}+1)}\, |n_{pX}+1\rangle\,, \\ \hat{P}_{pX} |n_{pX}\rangle &= \sqrt{(\Omega_{pX} - n_{pX}+1)n_{pX}}\, |n_{pX}-1\rangle\,. \end{aligned} \tag{3}$$

These factors retain Pauli blocking exactly within the chosen space.

For each isolated fragment, we use

$$\hat{H}_X = \sum_{p\in X} 2\epsilon_{pX} \hat{n}_{pX} - G_X \sum_{p,p'\in X} \hat{P}^{\dagger}_{pX} \hat{P}_{p'X}, \tag{4}$$

This is the constant-strength pairing Hamiltonian, where ϵ_{pX} is the single-neutron energy, $\hat{n}_{pX}$ is the pair-number operator, and $G_X = 20.5/A_X$ MeV is the intrafragment pairing strength for mass number A_X. The indices p and p' are summed independently. Thus each off-diagonal pair-scattering process $p' \to p$ appears together with its Hermitian conjugate $p \to p'$, and the diagonal contribution $\hat{P}^{\dagger}_{pX}\hat{P}_{pX}$ is retained. The prescription for G_X follows the standard finite-nucleus pairing systematics [2].

For a fixed number M_X of active pairs, diagonalization of Eq. (4) defines the normalized ground state of the isolated fragment,

$$\hat{H}_X\, |\mathrm{GS}_X(M_X)\rangle = E^{(0)}_X(M_X)\, |\mathrm{GS}_X(M_X)\rangle\,. \tag{5}$$

The superscript (0) labels the ground state and does not denote a noninteracting energy. Its mean pair occupation

TABLE I. Active neutron orbitals for ^{116}Sn (P) and ^{60}Ni (T). The listed ϵ_{pX} are single-neutron energies, Ω_{pX} is the maximum number of pairs, and $\langle\hat{n}_{pX}\rangle$ is the calculated mean pair occupation in the entrance ground state.

X	p	ϵ_{pX} (MeV)	Ω_{pX}	$\langle\hat{n}_{pX}\rangle$
T	$2p_{3/2}$	-11.718	2	1.791249
	$2p_{1/2}$	-10.048	1	0.057761
	$1f_{5/2}$	-9.718	3	0.126061
	$1g_{9/2}$	-5.548	5	0.024930
P	$1g_{7/2}$	-10.860	4	3.685572
	$3s_{1/2}$	-9.700	1	0.479878
	$2d_{3/2}$	-9.510	2	0.577915
	$1h_{11/2}$	-8.120	6	0.237308
	$2f_{7/2}$	-3.110	4	0.014270
	$3p_{3/2}$	-1.770	2	0.005057

in orbital p is

$$\langle\hat{n}_{pX}\rangle \equiv \langle \mathrm{GS}_X(M_X)|\,\hat{n}_{pX}\,|\mathrm{GS}_X(M_X)\rangle\,. \tag{6}$$

For the entrance channel, $M_P = 5$ and $M_T = 2$; hence the target occupations in Table I are $\langle\hat{n}_{pT}\rangle = \langle \mathrm{GS}_T(2)|\,\hat{n}_{pT}\,|\mathrm{GS}_T(2)\rangle$.

Table I gives the space used for ^{116}Sn+^{60}Ni. The filled $1g_{9/2}$ and $2d_{5/2}$ orbitals of ^{116}Sn are frozen, leaving $M_P = 5$ active pairs, while ^{60}Ni contains $M_T = 2$. The last column gives the occupations defined in Eq. (6). Their unrounded sums are $\sum_{p\in P}\langle\hat{n}_{pP}\rangle = 5$ and $\sum_{p\in T}\langle\hat{n}_{pT}\rangle = 2$. They count pairs rather than individual neutrons; the fill of each pair slot in Fig. 1(b) is $\langle\hat{n}_{pX}\rangle/\Omega_{pX}$. A common shift $\delta\epsilon_T = -0.387763$ MeV is included in the target energies shown in the table. It aligns the theoretical ground-state-to-ground-state energy release with the measured value [6],

$$E_P^{(0)}(5)+E_T^{(0)}(2)-E_P^{(0)}(4)-E_T^{(0)}(3) = Q_{2n}^{\mathrm{exp}} = 1.30\ \mathrm{MeV}. \tag{7}$$

B. Collision dynamics

The two fragments are coupled by a term that moves one correlated pair at a time,

$$\begin{aligned}\hat{H}(t,D) &= \hat{H}_P + \hat{H}_T + \hat{H}_{PT}(t,D),\\ \hat{H}_{PT}(t,D) &= -G_{PT}(t,D)\sum_{\substack{p\in P\\ q\in T}}\left(\hat{P}^\dagger_{qT}\hat{P}_{pP} + \hat{P}^\dagger_{pP}\hat{P}_{qT}\right).\end{aligned} \tag{8}$$

The real scalar G_{PT} is an effective interfragment pairing strength. The first term in parentheses transfers a pair from projectile to target and the second describes the reverse process; hence Eq. (8) is Hermitian and conserves the total pair number $M = M_P + M_T$. We characterize each collision by the center-to-center distance of closest approach of its repulsive Rutherford trajectory. For a projectile P incident on a stationary target T, this distance may be written in either center-of-mass or laboratory variables as

$$\begin{aligned}D &= \frac{Z_P Z_T e^2}{2E_{\mathrm{cm}}}\left[1+\csc\left(\frac{\theta_{\mathrm{cm}}}{2}\right)\right]\\ &= \frac{Z_P Z_T e^2 (A_P+A_T)}{2E_{\mathrm{lab}}A_T}\left(1+\frac{1}{\cos\theta_{\mathrm{lab}}}\right).\end{aligned} \tag{9}$$

Here E_{cm} and θ_{cm} are the asymptotic center-of-mass energy and scattering angle. The second form uses $E_{\mathrm{cm}} = E_{\mathrm{lab}}A_T/(A_P+A_T)$ and applies when target-like recoils are detected at the laboratory angle θ_{lab}, for which $\sin(\theta_{\mathrm{cm}}/2) = \cos\theta_{\mathrm{lab}}$. It is the convention used to construct the experimental abscissa in Ref. [6]. Nuclear deflection is not included in this mapping.

For a collision with the closest approach D, we take

$$\begin{aligned}G_{PT}(t,D) &= G_{\mathrm{pk}}(D)\exp\left[-\left(\frac{t}{\tau(D)}\right)^2\right],\\ G_{\mathrm{pk}}(D) &= \frac{g_0}{1+\exp[(D-D_B)/a_g]}.\end{aligned} \tag{10}$$

Time $t = 0$ is the turning point, G_{pk} is the peak coupling, g_0 is its interior limiting value, D_B is a reference barrier distance, and a_g controls the radial falloff. We use $D_B = 12.13$ fm for ^{116}Sn + ^{60}Ni and $D_B = 11.47$ fm for ^{96}Zr + ^{40}Ca. In each case D_B is held fixed while g_0 and a_g are calibrated to the corresponding transfer data in Sec. III; the parameters are collected in Table II. The pulse duration is estimated from the local curvature of a head-on Coulomb trajectory,

$$\begin{aligned}\tau(D) &= \frac{1}{c}\sqrt{\frac{2a_{\mathrm{opt}}}{\alpha_C(D)}},\\ \alpha_C(D) &= \frac{Z_P Z_T e^2}{\mu_R c^2 D^2},\\ \mu_R &= \frac{A_P A_T}{A_P+A_T}m_u.\end{aligned} \tag{11}$$

Here α_C is the Coulomb curvature expressed as an inverse length, Z_X is the proton number, μ_R is the reduced mass, m_u is the atomic mass unit, $e^2 = 1.439964$ MeV fm, and c is the speed of light. We identify a_{opt} with the surface diffuseness of the Broglia–Winther ion–ion potential [7],

$$a_{\mathrm{opt}}^{-1} = 1.17\left[1+0.53\left(A_P^{-1/3}+A_T^{-1/3}\right)\right]\ \mathrm{fm}^{-1}. \tag{12}$$

This mass-dependent prescription gives $a_{\mathrm{opt}} = 0.687$ fm for ^{116}Sn+^{60}Ni and 0.673 fm for ^{96}Zr+^{40}Ca. The former is also the Woods–Saxon diffuseness quoted explicitly in the reaction analysis of Ref. [6], while the ^{96}Zr + ^{40}Ca analysis uses the Broglia–Winther empirical ion–ion potential [5]. These values are not fitted to the pair-transfer probabilities: they set only the time scale $\tau(D)$, whereas the radial range a_g of the transfer coupling is calibrated separately. The derivation and assumptions of the duration estimate are given in Appendix A.

For a fixed D, the entrance state is the product of the isolated-fragment ground states defined in Eq. (5). Its evolution is

$$|\Psi(t_i, D)\rangle = |\mathrm{GS}_T(2)\rangle \otimes |\mathrm{GS}_P(5)\rangle\,,$$
$$\mathrm{i}\hbar\frac{\partial}{\partial t}|\Psi(t, D)\rangle = \hat{H}(t, D)\,|\Psi(t, D)\rangle\,, \tag{13}$$

where $\hbar$ is the reduced Planck constant. We propagate from $t_i = -5\tau(D)$ to $t_f = 5\tau(D)$; at these endpoints the Gaussian coupling is smaller than its peak by e^{-25}. Evolution takes place in the $M = 7$ sector.

Let $d_X(M_X)$ denote the number of seniority-zero configurations of fragment X with M_X pairs. We generate them by enumerating all integer occupations that satisfy

$$\sum_{p=1}^{L_X} n_{pX} = M_X,$$
$$n_{pX} \in \mathbb{Z}, \quad 0 \le n_{pX} \le \Omega_{pX}, \quad p = 1, \ldots, L_X. \tag{14}$$

In practice, one loops over $n_{pX} = 0, 1, \ldots, \Omega_{pX}$ for every orbital and retains a tuple only when its entries sum to M_X. Each retained tuple is one basis state, so their number is $d_X(M_X)$. For the entrance projectile, $M_P = 5$ and the capacities are $(4, 1, 2, 6, 4, 2)$; direct enumeration gives $d_P(5) = 154$. For the target, $M_T = 2$ and the capacities are $(2, 1, 3, 5)$; the same enumeration gives $d_T(2) = 9$. Applying this procedure to every partition of the seven pairs gives the full fixed-number dimension,

$$\begin{aligned}\dim \mathcal{H}_{M=7} &= \sum_{M_T=0}^{7} d_T(M_T) d_P(7 - M_T) \\ &= (1\times 295) + (4\times 225) + (9\times 154) + (15\times 93) \\ &\quad + (20\times 48) + (23\times 20) + (23\times 6) + (20\times 1) \\ &= 5554.\end{aligned} \tag{15}$$

The two factors in each product are, respectively, the numbers of target and projectile configurations for $M_T = 0, 1, \ldots, 7$ and $M_P = 7 - M_T$. Thus the entrance partition $(M_T, M_P) = (2, 5)$ contains $9 \times 154 = 1386$ product states, while the one-pair pickup partition $(3, 4)$ contains $15 \times 93 = 1395$. Summing all allowed pair partitions gives the complete seniority-zero basis of the chosen active space.

C. Matrix elements in the occupation basis

The numerical Hamiltonian is constructed in the orthonormal product basis $|\boldsymbol{n}\rangle = |\boldsymbol{n}_T\rangle \otimes |\boldsymbol{n}_P\rangle$. It is useful to denote the raising and lowering amplitudes in Eq. (3) by

$$A^+_{pX}(n) = \sqrt{(\Omega_{pX} - n)(n + 1)},$$
$$A^-_{pX}(n) = \sqrt{n(\Omega_{pX} - n + 1)}. \tag{16}$$

For $p = p'$, the lowering and raising operations return the same configuration. The resulting diagonal matrix element is

$$\begin{aligned}\langle \boldsymbol{n}|\,\hat{H}\,|\boldsymbol{n}\rangle = &\sum_{X=P,T}\sum_{p\in X} 2\epsilon_{pX} n_{pX} \\ &- \sum_{X=P,T} G_X \sum_{p\in X} n_{pX}(\Omega_{pX} - n_{pX} + 1).\end{aligned} \tag{17}$$

The transfer interaction has no diagonal contribution.

Let $\boldsymbol{e}_{pX}$ increase the occupation of orbital p in fragment X by one. For two different orbitals in the same fragment, the term $\hat{P}^\dagger_{pX}\hat{P}_{p'X}$ moves one pair from p' to p. Its only matrix element from $|\boldsymbol{n}\rangle$ is

$$\langle \boldsymbol{n}'|\,\hat{H}_X\,|\boldsymbol{n}\rangle = -G_X A^+_{pX}(n_{pX}) A^-_{p'X}(n_{p'X}), \tag{18}$$

where $\boldsymbol{n}' = \boldsymbol{n} + \boldsymbol{e}_{pX} - \boldsymbol{e}_{p'X}$ and $p \neq p'$. Similarly, the two directions of interfragment transfer give

$$\begin{aligned}\langle \boldsymbol{n}'_{P\to T}|\,\hat{H}_{PT}\,|\boldsymbol{n}\rangle &= -G_{PT}(t, D) A^+_{qT}(n_{qT}) A^-_{pP}(n_{pP}), \\ \langle \boldsymbol{n}'_{T\to P}|\,\hat{H}_{PT}\,|\boldsymbol{n}\rangle &= -G_{PT}(t, D) A^+_{pP}(n_{pP}) A^-_{qT}(n_{qT}),\end{aligned} \tag{19}$$

where $\boldsymbol{n}'_{P\to T} = \boldsymbol{n} + \boldsymbol{e}_{qT} - \boldsymbol{e}_{pP}$ and $\boldsymbol{n}'_{T\to P} = \boldsymbol{n} + \boldsymbol{e}_{pP} - \boldsymbol{e}_{qT}$. A move is allowed only when its source orbital is occupied and its destination orbital is not full; otherwise the corresponding amplitude in Eq. (16) vanishes. All other off-diagonal elements are zero. The reverse move gives the transposed matrix element, so the Hamiltonian matrix is real and symmetric.

D. Pair transfer and entanglement observables

Let $|\alpha\rangle = |\boldsymbol{n}_T(\alpha)\rangle \otimes |\boldsymbol{n}_P(\alpha)\rangle$ denote a product-basis state and $M_T(\alpha) = \sum_{q\in T} n_{qT}(\alpha)$ its target pair number. The projector onto net transfer κ to the target and its final probability are

$$\hat{\Pi}_\kappa = \sum_{\alpha:\, M_T(\alpha) = 2+\kappa} |\alpha\rangle\langle\alpha|\,,$$
$$\mathcal{P}_\kappa(D) = \langle\Psi(t_f, D)|\,\hat{\Pi}_\kappa\,|\Psi(t_f, D)\rangle\,. \tag{20}$$

Thus $\kappa = +1$ corresponds to $(M_T, M_P) = (3, 4)$, i.e., the reaction $^{116}\mathrm{Sn} + {}^{60}\mathrm{Ni} \to {}^{114}\mathrm{Sn} + {}^{62}\mathrm{Ni}$. The quantity compared with the experimental transfer yield is the ratio

$$P_{+1}(D) = \frac{\mathcal{P}_{+1}(D)}{\mathcal{P}_0(D)}, \tag{21}$$

not the normalized sector probability $\mathcal{P}_{+1}$ alone. Other transfer channels are obtained by changing κ.

The same wave function also resolves correlations generated during the collision. We quantify the entanglement between target and projectile by

$$\rho_T(t, D) = \mathrm{Tr}_P\,|\Psi(t, D)\rangle\langle\Psi(t, D)|\,,$$
$$\mathcal{S}_{T|P}(t, D) = -\,\mathrm{Tr}_T\left[\rho_T(t, D)\log_2\rho_T(t, D)\right], \tag{22}$$

where Tr_P is the partial trace over projectile states and $\mathcal{S}_{T|P}$ is the base-2 von Neumann entropy, measured in bits.

E. Qubit representation

For a direct qubit implementation, resolve each orbital p into Ω_{pX} hard-core pair slots $i = (X, p, \lambda)$, with $\lambda = 1, \ldots, \Omega_{pX}$. If $\hat{b}_i^\dagger$ fills slot i and $\hat{n}_i = \hat{b}_i^\dagger \hat{b}_i$, then, in the permutation-symmetric subspace of each orbital,

$$\hat{P}_{pX}^\dagger = \sum_{\lambda=1}^{\Omega_{pX}} \hat{b}_{Xp\lambda}^\dagger, \qquad \hat{n}_{pX} = \sum_{\lambda=1}^{\Omega_{pX}} \hat{n}_{Xp\lambda}. \tag{23}$$

This identity reproduces Eq. (3), as derived in Appendix B; it is therefore an exact representation of the seniority-zero model, not a bosonic approximation. We encode an empty (occupied) slot as $|0_i\rangle$ ($|1_i\rangle$), choose $Z_i |0_i\rangle = + |0_i\rangle$, and use

$$\hat{b}_i^\dagger = \frac{X_i - \mathrm{i}Y_i}{2}, \qquad \hat{b}_i = \frac{X_i + \mathrm{i}Y_i}{2}, \qquad \hat{n}_i = \frac{\hat{I} - Z_i}{2}, \tag{24}$$

where X_i, Y_i, and Z_i are Pauli operators, $\hat{I}$ is the identity, and $\mathrm{i}^2 = -1$.

For two slots $i = (X, p, \lambda)$ and $j = (Y, q, \mu)$, where $X, Y \in \{P, T\}$ label the fragments, let $\epsilon_i = \epsilon_{pX}$. The symmetric coupling matrix is

$$G_{ij}(t, D) = \begin{cases} G_P, & X = Y = P, \\ G_T, & X = Y = T, \\ G_{PT}(t, D), & X \neq Y. \end{cases} \tag{25}$$

Equations (4) and (8) then become

$$\hat{H}(t, D) = E_c \hat{I} + \sum_i \zeta_i Z_i + \sum_{i<j} J_{ij}(t, D)(X_i X_j + Y_i Y_j), \tag{26}$$

with

$$\begin{aligned} \zeta_i &= -\epsilon_i + \frac{G_{ii}}{2}, \\ J_{ij}(t, D) &= -\frac{G_{ij}(t, D)}{2}, \\ E_c &= \sum_i \left(\epsilon_i - \frac{G_{ii}}{2} \right). \end{aligned} \tag{27}$$

The scalar E_c contributes only a global phase. The capacities in Table I give $N_q = 11 + 19 = 30$ pair slots and hence 30 qubits, of which seven are occupied initially. Their conserved occupation is

$$\hat{M} = \sum_{i=1}^{N_q} \hat{n}_i = \frac{N_q \hat{I} - \sum_i Z_i}{2}, \qquad [\hat{H}, \hat{M}] = 0. \tag{28}$$

The model thus maps exactly to longitudinal fields and number-conserving XY exchange couplings. It retains finite-number pairing, Pauli blocking, coherent multi-orbital transfer, and interfragment entanglement, while the single scalar $G_{PT}(t, D)$ replaces orbital-dependent transfer form factors. Broken pairs, absorption, continuum channels, and quantized relative motion are outside the present model space; reproducing Eq. (26) on hardware therefore tests the simulator independently of these reaction-model limitations.

TABLE II. Reaction-dependent barrier distances and calibration of the transfer coupling. The values of D_B are held fixed, whereas g_0 and a_g are fitted separately for the two reactions. The last column gives the geometric rms factor between the calculated and measured transfer probabilities.

System	D_B (fm)	g_0 (MeV)	a_g (fm)	f_{rms}
^{116}Sn + ^{60}Ni	12.13	0.1603	0.6240	1.273
^{96}Zr + ^{40}Ca	11.47	0.3010	0.7294	1.214

III. NUMERICAL RESULTS

A. Details of the calculations

We solve Eq. (13) by evolving every coefficient in the complete seniority-zero space at fixed total pair number, with the Hamiltonian applied through sparse matrix–vector products. No instantaneous diagonalization of the time-dependent Hamiltonian is performed. We refer to this procedure as direct sparse propagation; its implementation is detailed in Appendix C. The isolated fragment ground states are obtained by exact diagonalization, and the collision state is propagated with an eighth-order adaptive integrator from $-5\tau(D)$ to $5\tau(D)$. The ^{116}Sn + ^{60}Ni calculation has the 5554-dimensional basis specified in Sec. II. For ^{96}Zr + ^{40}Ca, we retain five upper-shell orbitals of ^{96}Zr containing three active pairs and four initially empty orbitals above the $N = 20$ closure of ^{40}Ca. Their combined fixed-number space has dimension 145 and maps onto 26 pair-slot qubits. We use $G_{\mathrm{Zr}} = 0.218$ MeV and $G_{\mathrm{Ca}} = 0.5125$ MeV, with the single-particle inputs of Ref. [5]. The corresponding barrier distance is $D_B = 11.47$ fm; its surface-overlap length $a_{\mathrm{opt}} = 0.673$ fm follows from Eq. (12).

With D_B fixed for each entrance channel, g_0 and a_g are calibrated separately for each reaction by minimizing unweighted residuals in $\log_{10} P_{2n}$. The data points were digitized from the published figures in Refs. [5, 6]. Table II summarizes the fixed barrier anchors and calibrated parameters. For ^{116}Sn + ^{60}Ni, the shifted target spectrum of Table I gives the measured ground-state Q value through Eq. (7). The ^{96}Zr + ^{40}Ca data are inclusive in the $+2n$ channel and contain unresolved excited-state strength. We therefore place the centroid of the retained $+1$-pair sector at the optimum value $Q = 0$, close to the approximately -0.23 MeV channel associated with the dominant calculated 0^+ strength in ^{42}Ca [5]. This is an explicit inclusive-channel approximation and must not be interpreted as the ground-state reaction value $Q_{gg} = +5.53$ MeV.

At the reference distances used below, repeating the calculations with tighter integration tolerances changes the transfer probabilities by less than 1.4×10^{-17} and the entanglement entropy by less than 7×10^{-12} bits for ^{116}Sn + ^{60}Ni. The corresponding changes for ^{96}Zr + ^{40}Ca are below 6×10^{-18} and 3×10^{-17} bits. The propagated norms remain within 9×10^{-15} of unity. These checks establish numerical convergence within the stated finite

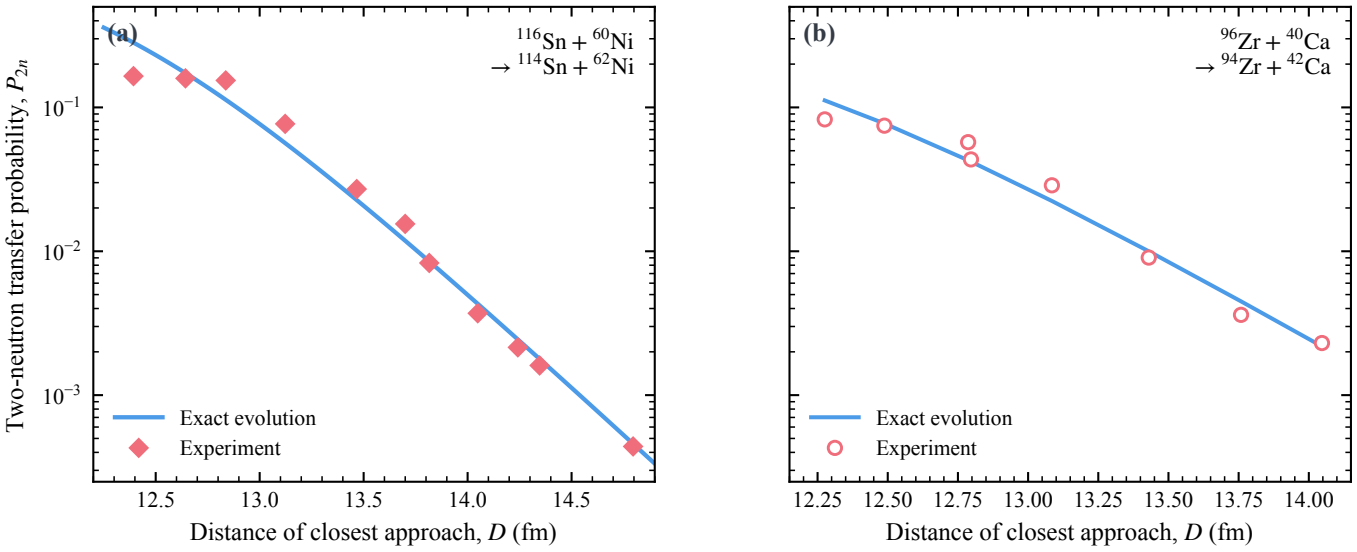


FIG. 2. Two-neutron transfer probability as a function of the distance of closest approach. (a) ^{116}Sn+^{60}Ni → ^{114}Sn+^{62}Ni; the solid curve shows the calculated ratio $\mathcal{P}_{+1}/\mathcal{P}_0$, and the filled diamonds are digitized from Ref. [6]. (b) ^{96}Zr+^{40}Ca → ^{94}Zr+^{42}Ca; the solid curve connects the calculated $\mathcal{P}_{+1}$ values, and the open circles are the inclusive $+2n$ data digitized from Ref. [5]. The calculations use the reaction-specific fixed values of D_B in Table II.

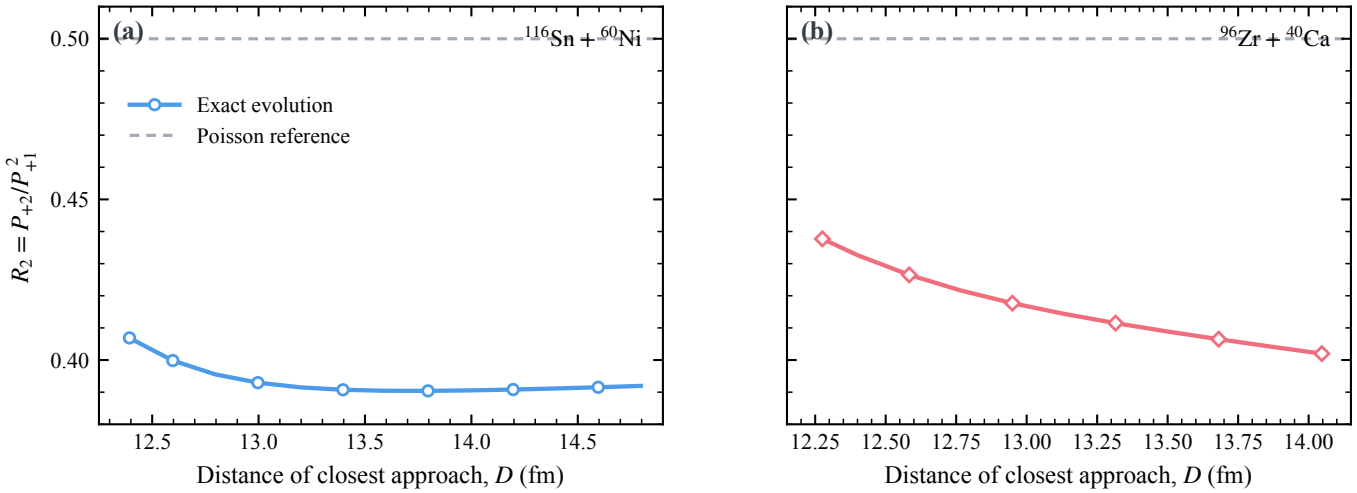


FIG. 3. Ratio $R_2(D) = P_{+2}(D)/[P_{+1}(D)]^2$ for (a) ^{116}Sn + ^{60}Ni and (b) ^{96}Zr + ^{40}Ca. Here $P_{+\kappa} = \mathcal{P}_{+\kappa}/\mathcal{P}_0$ follows the normalization of Eq. (21). The dashed line at $R_2 = 1/2$ is the Poisson value for independent rare one-pair transfers and is shown only as a reference. Markers select calculated distances, and the curves connect the exact results.

spaces; they do not estimate the systematic uncertainty from the model-space truncation or the assumed radial coupling.

B. Transfer probabilities

Figure 2 compares the exact finite-space evolution with the measured distance dependence. For ^{116}Sn + ^{60}Ni → ^{114}Sn + ^{62}Ni, the plotted observable is the ratio $\mathcal{P}_{+1}/\mathcal{P}_0$ defined in Eq. (21). Two coupling parameters reproduce the decrease of the probability over nearly three orders of magnitude. The geometric rms mismatch is a factor of 1.273, while the largest pointwise discrepancy is a factor of 1.71 at the smallest distance. The fitted curve should be viewed as a calibrated description of the radial trend rather than a parameter-free prediction: $G_{PT}(t, D)$ absorbs transfer form factors, trajectory averaging, and an overall reaction normalization that are not calculated in the reduced model.

The same construction describes ^{96}Zr + ^{40}Ca → ^{94}Zr + ^{42}Ca with a geometric rms factor of 1.214 and a maximum discrepancy of 1.35. Here $\mathcal{P}_{+1}$ is compared directly with the inclusive $+2n$ probability because the entrance ^{40}Ca active space is empty. The two experimental values at $D = 12.7863$ and 12.7965 fm differ appreciably, whereas a model depending on the collision only through D necessarily assigns them nearly equal probabilities.

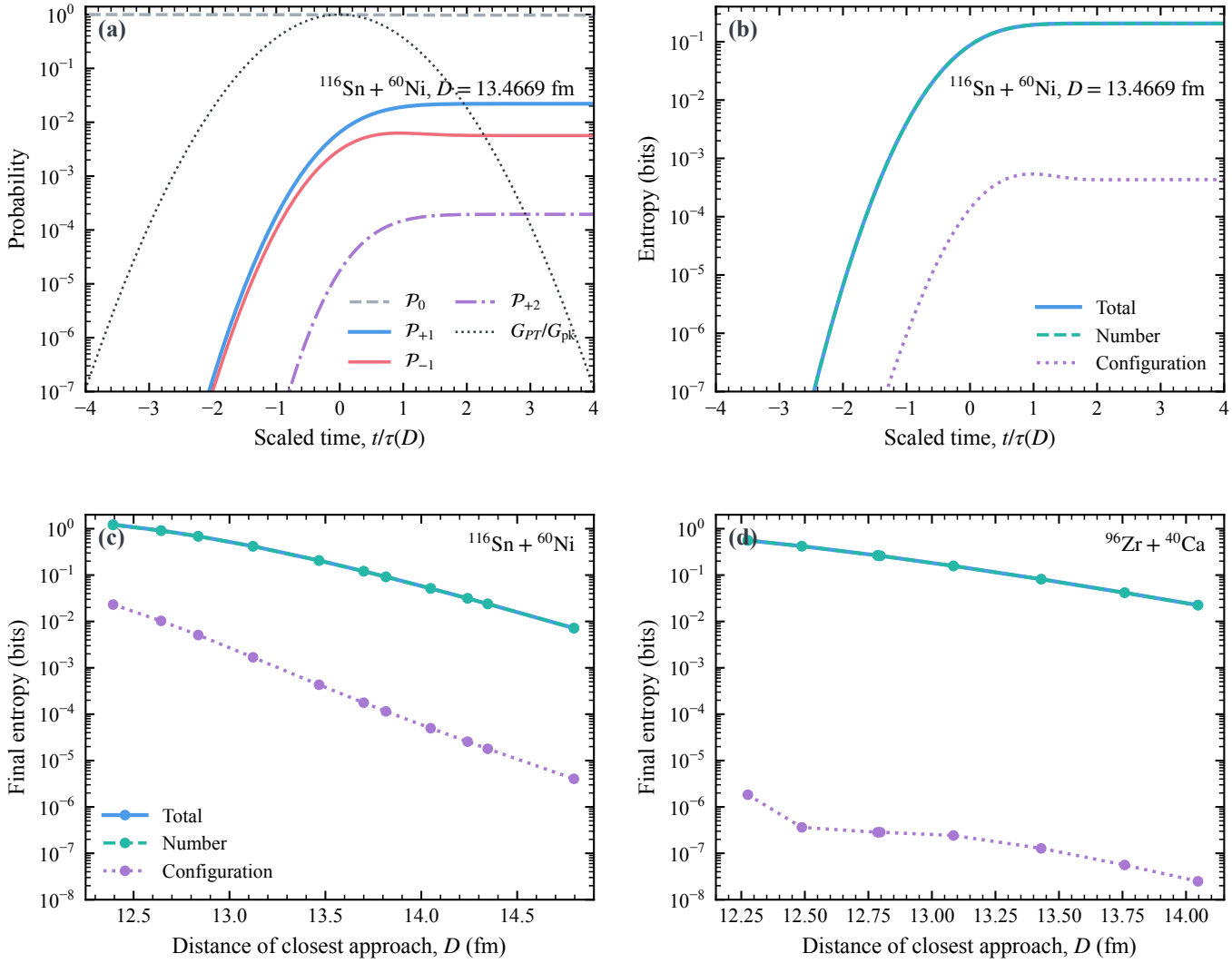


FIG. 4. Number-resolved dynamics and interfragment entanglement. (a) Transfer-sector probabilities for ^{116}Sn + ^{60}Ni at $D = 13.4669$ fm; the normalized coupling pulse is included to set the time scale. (b) Total, number, and configuration entropies for the same collision. (c),(d) Final entropy components as functions of D for ^{116}Sn + ^{60}Ni and ^{96}Zr + ^{40}Ca, respectively. The total and number curves nearly coincide; the logarithmic scale resolves their difference $\mathcal{S}_{\rm cfg}$. Lines in panels (c) and (d) connect the calculated distances.

This local mismatch exposes information carried by the beam energy, scattering angle, and unresolved final-state distribution that is lost when the relative motion is reduced to a single closest-approach coordinate. The overall agreement therefore supports the use of the model as a compact benchmark, but it does not turn its sector probability into an absolute reaction cross section.

The +2 sector represents transfer of two neutron pairs, or four neutrons. To test whether it follows from statistically independent repetitions of the one-pair channel, we define

$$R_2(D) = \frac{P_{+2}(D)}{[P_{+1}(D)]^2} = \frac{\mathcal{P}_{+2}(D)\mathcal{P}_0(D)}{[\mathcal{P}_{+1}(D)]^2}. \qquad (29)$$

Independent rare events with a Poisson distribution give $R_2 = 1/2$, as derived in Appendix D. Figure 3 shows that the exact finite-space results remain below this value throughout the displayed interval. For ^{116}Sn + ^{60}Ni, R_2 stays within 0.390–0.407, with its largest value at $D = 12.3932$ fm. For ^{96}Zr + ^{40}Ca, it increases steadily as the nuclei approach, from 0.402 at $D = 14.0468$ fm to 0.438 at $D = 12.2756$ fm. Thus P_{+2} retains the leading P_{+1}^2 radial scaling, but its coefficient is neither the Poisson value nor universal between the two active spaces. The sub-Poisson coefficient is generated by a coherent finite-space evolution in which level mismatch, Pauli blocking, and interference among multiorbital pathways coexist. Since these ingredients have not been varied independently, the deviation cannot be assigned uniquely to any one of them. The moderate increase toward smaller D marks departures from lowest-order scaling as the coupling pulse grows. Finally, R_2 is a model diagnostic rather than an experimental enhancement factor because four-neutron transfer data were not

used in the calibration.

C. Entanglement generated by pair transfer

Because the total pair number is fixed, the reduced density matrix of either fragment is block diagonal in its local pair number. The entropy in Eq. (22) consequently has the exact decomposition

$$\mathcal{S}_{T|P} = \mathcal{S}_{\rm num} + \mathcal{S}_{\rm cfg},$$
$$\mathcal{S}_{\rm num} = -\sum_{\kappa} \mathcal{P}_{\kappa} \log_2 \mathcal{P}_{\kappa}. \tag{30}$$

Here $\mathcal{S}_{\rm num}$ is the number entropy associated with fluctuations in the pair partition between the fragments, whereas $\mathcal{S}_{\rm cfg}$ is the configurational entropy generated by entanglement among orbital configurations within each fixed-number sector. This decomposition separates correlations arising from pair-number transfer from those internal to a transfer sector.

Figure 4(a) follows the number-resolved dynamics of ^{116}Sn + ^{60}Ni at $D = 13.4669$ fm. After the coupling pulse has vanished, the dominant nonelastic sector has $\mathcal{P}_{+1} = 0.02206$, while $\mathcal{P}_{-1} = 0.00566$ and $\mathcal{P}_{+2} = 1.96 \times 10^{-4}$. The elastic sector remains $\mathcal{P}_0 = 0.97209$, giving $\mathcal{P}_{+1}/\mathcal{P}_0 = 0.02269$. The finite reverse-transfer component follows from the Hermitian coupling in Eq. (8); it is not removed by imposing a one-way reaction channel. Panel (b) shows that the entropy rises during the same pulse and saturates at 0.20622 bits. Of this value, 0.20579 bits is number entropy and 4.31×10^{-4} bits is configuration entropy. Thus, at this distance, more than 99.7% of the final interfragment entanglement records the coherent superposition of different transfer sectors.

The distance systematics in Figs. 4(c) and 4(d) follow the rapid radial attenuation of the transfer coupling. For ^{116}Sn + ^{60}Ni, the total final entropy decreases from 1.2255 bits at $D = 12.3932$ fm to 7.17×10^{-3} bits at $D = 14.7959$ fm. The configuration contribution is largest at the smallest distance, where repeated and reverse transfer have sufficient amplitude to mix multiple configurations within a number sector, but it remains only 2.31×10^{-2} bits. For ^{96}Zr + ^{40}Ca, the total entropy decreases from 0.55985 bits at $D = 12.2756$ fm to 0.02249 bits at $D = 14.0468$ fm. Its configuration component is below 1.9×10^{-6} bits throughout the displayed range. At the reference distance $D = 13.4298$ fm, the final distribution has $\mathcal{P}_0 = 0.989962$, $\mathcal{P}_{+1} = 0.009996$, and $\mathcal{P}_{+2} = 4.14 \times 10^{-5}$, producing $\mathcal{S}_{T|P} = 0.08143$ bits.

The dominance of $\mathcal{S}_{\rm num}$ in both reactions gives the entanglement a simple physical interpretation: within the present active spaces, a sub-barrier encounter primarily creates a coherent superposition of fragments differing by whole neutron pairs. The much smaller $\mathcal{S}_{\rm cfg}$ quantifies additional multiorbital structure inside those transfer sectors. These entropies are properties of the closed, unitary two-fragment model and are not direct experimental observables. Absorption, broken pairs, continuum states, and coupling to unobserved reaction channels would redistribute or decohere them. Subject to that limitation, the converged probabilities and entropies provide exact classical reference values for testing a quantum-hardware implementation of Eq. (26).

IV. CONCLUSIONS

We have introduced a number-conserving two-fragment pairing model for coherent neutron-pair transfer at sub-barrier energies. Each nucleus is represented by a finite seniority-zero pairing space, and the collision is reduced to a time-dependent Hermitian coupling that exchanges $J = 0$ pairs between the fragments. The resulting quasi-spin Hamiltonian retains finite-space pairing, Pauli blocking, multiorbital coherence, and interfragment entanglement, while mapping exactly onto longitudinal fields and number-conserving XY couplings in a pair-slot qubit representation.

After calibrating the two parameters governing the radial transfer coupling separately for each reaction, the model captures the measured distance dependence of ^{116}Sn+^{60}Ni $\rightarrow$ ^{114}Sn+^{62}Ni and ^{96}Zr+^{40}Ca $\rightarrow$ ^{94}Zr+^{42}Ca with geometric rms factors of 1.273 and 1.214, respectively. The calculated two-pair ratio P_{+2}/P_{+1}^2 remains below the independent-transfer Poisson value, demonstrating that its coefficient is shaped by the coherent finite-space dynamics rather than by repeated statistically independent events. Pair transfer also generates interfragment entanglement. Its dominant contribution is the number entropy associated with a coherent superposition of different pair partitions, whereas the configurational entropy within a fixed transfer sector is much smaller for the cases studied here.

The present construction provides a controlled many-body benchmark that is both classically verifiable and directly implementable on near-term quantum hardware. Classical propagation supplies reference values for the complete transfer distribution and interfragment entanglement, enabling quantitative validation of a quantum implementation. From this foundation, larger active spaces, seniority-changing configurations, orbital-dependent interfragment couplings, continuum and absorptive channels, and a dynamical treatment of the relative motion can be incorporated progressively. These extensions offer a systematic route from the compact model developed here toward increasingly realistic quantum simulations of nuclear transfer reactions.

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Sol for assistance with code development, mathematical derivations, figure preparation (writing Python scripts to generate the data plots; no AI-generated images were used), and language editing. The AI-generated outputs were independently verified and audited by the authors. The authors take full responsibility for the scientific content of the manuscript.

Appendix A: Derivation of the Coulomb-curvature collision time

Equation (11) follows by expanding a head-on Coulomb trajectory about its turning point and assuming an exponential radial form factor for pair transfer. The latter is a modeling choice, not a consequence of Rutherford mechanics.

Let $\boldsymbol{r}_P$ and $\boldsymbol{r}_T$ be the fragment coordinates and $\boldsymbol{R} = \boldsymbol{r}_P - \boldsymbol{r}_T$ their relative coordinate. In the center-of-mass frame, $m_P\boldsymbol{r}_P + m_T\boldsymbol{r}_T = 0$, and hence

$$\boldsymbol{r}_P = \frac{m_T}{m_P + m_T}\boldsymbol{R}, \qquad \boldsymbol{r}_T = -\frac{m_P}{m_P + m_T}\boldsymbol{R}. \tag{A1}$$

Substitution into the two-body kinetic energy gives

$$\begin{aligned} T &= \frac{1}{2}m_P\dot{\boldsymbol{r}}_P^2 + \frac{1}{2}m_T\dot{\boldsymbol{r}}_T^2 = \frac{1}{2}\mu_R\dot{\boldsymbol{R}}^2, \\ \mu_R &= \frac{m_P m_T}{m_P + m_T}. \end{aligned} \tag{A2}$$

Approximating the fragment masses by $m_X = A_X m_u$ immediately yields the last line of Eq. (11),

$$\mu_R = \frac{A_P A_T}{A_P + A_T}m_u. \tag{A3}$$

For a head-on trajectory the relative motion is radial. Outside the nuclear-interaction region its potential is

$$V_C(R) = \frac{K_C}{R}, \qquad K_C = Z_P Z_T e^2, \tag{A4}$$

where e^2 includes the Coulomb constant. Newton's equation for the relative coordinate is therefore

$$\mu_R\ddot{R} = -\frac{dV_C}{dR} = \frac{K_C}{R^2}. \tag{A5}$$

Choose $t = 0$ at the turning point. Then $R(0) = D$ and $\dot{R}(0) = 0$, while Eq. (A5) gives

$$\ddot{R}(0) = \frac{Z_P Z_T e^2}{\mu_R D^2}. \tag{A6}$$

The incoming and outgoing branches are related by time reversal, so $R(t)$ is even about $t = 0$. Its local expansion is consequently

$$\begin{aligned} R(t) - D &= \frac{1}{2}\ddot{R}(0)t^2 + \mathcal{O}(t^4) \\ &= \frac{1}{2}\alpha_C(D)(ct)^2 + \mathcal{O}(t^4), \end{aligned} \tag{A7}$$

with

$$\alpha_C(D) \equiv \frac{\ddot{R}(0)}{c^2} = \frac{Z_P Z_T e^2}{\mu_R c^2 D^2}. \tag{A8}$$

Thus α_C is the outward radial acceleration in the length coordinate ct and has units of inverse length.

The trajectory alone does not determine the time dependence of the transfer matrix element. We additionally assume that, near closest approach, its radial form factor decreases exponentially as the surfaces separate,

$$\frac{G_{PT}[R(t), D]}{G_{\rm pk}(D)} \simeq \exp\left[-\frac{R(t) - D}{a_{\rm opt}}\right]. \tag{A9}$$

The length $a_{\rm opt}$ is adopted from the mass-dependent ion–ion surface diffuseness in Eq. (12). This connects the duration estimate to a standard reaction-potential length scale, but it does not imply that the pair-transfer form factor is identical to the optical potential. In particular, $a_{\rm opt}$ is held fixed in $\tau(D)$, while a_g in Eq. (10) is fitted independently. Inserting Eq. (A7) into Eq. (A9) gives, to quadratic order about the turning point,

$$\frac{G_{PT}(t, D)}{G_{\rm pk}(D)} \simeq \exp\left[-\frac{\alpha_C(D)c^2t^2}{2a_{\rm opt}}\right]. \tag{A10}$$

Comparison with the Gaussian definition $G_{PT}/G_{\rm pk} = \exp[-(t/\tau)^2]$ yields

$$\tau(D) = \frac{1}{c}\sqrt{\frac{2a_{\rm opt}}{\alpha_C(D)}} = \frac{D}{c}\sqrt{\frac{2a_{\rm opt}\mu_R c^2}{Z_P Z_T e^2}}, \tag{A11}$$

which is the first line of Eq. (11). In nuclear units, e^2 has units MeV fm and $\mu_R c^2$ has units MeV, so α_C has units fm^{-1}. The square root in Eq. (A11) is therefore a length, and division by c gives a time.

As a numerical check, for ^{116}Sn + ^{60}Ni at $D = 13.4669$ fm, $a_{\rm opt} = 0.687$ fm gives $\mu_R c^2 = 3.6836 \times 10^4$ MeV, $\alpha_C = 3.0176 \times 10^{-4}$ fm^{-1}, and $c\tau = 67.48$ fm. Hence $\tau = 2.251 \times 10^{-22}$ s, the value used in the reference calculation.

Equation (A11) is a local time scale, not the exact duration of a general scattering orbit. Finite angular momentum adds a centrifugal contribution to the turning-point curvature, and a realistic trajectory also contains a nuclear potential and dissipative channels. The exponential form in Eq. (A9) is an additional modeling assumption rather than a consequence of Rutherford dynamics. These effects are omitted so that $\tau(D)$ depends only on D and fixed entrance-channel quantities; their influence is absorbed partly into the fitted $G_{\rm pk}(D)$.

Appendix B: Exact pair-slot representation of the seniority-zero space

We derive Eq. (23) for one orbital and suppress the labels pX. Let its pair degeneracy be Ω. Each slot

$\lambda = 1, \ldots, \Omega$ corresponds to one time-reversed doublet $(m_\lambda, -m_\lambda)$, and its pair operator is

$$\hat{b}^\dagger_\lambda = (-1)^{j-m_\lambda} \hat{a}^\dagger_{m_\lambda} \hat{a}^\dagger_{-m_\lambda}. \tag{B1}$$

Within the seniority-zero space, a doublet is either empty or occupied by one pair. Thus

$$\begin{aligned} \hat{b}^\dagger_\lambda |0_\lambda\rangle &= |1_\lambda\rangle, \qquad \hat{b}^\dagger_\lambda |1_\lambda\rangle = 0, \\ \hat{b}_\lambda |1_\lambda\rangle &= |0_\lambda\rangle, \qquad \hat{b}_\lambda |0_\lambda\rangle = 0. \end{aligned} \tag{B2}$$

Consequently $(\hat{b}^\dagger_\lambda)^2 = \hat{b}^2_\lambda = 0$ and $\{\hat{b}_\lambda, \hat{b}^\dagger_\lambda\} = \hat{I}$. Pair operators belonging to different doublets commute because each contains an even number of fermion operators. These are hard-core, rather than canonical boson, operators.

Let $|\mathrm{vac}\rangle$ denote the state with all Ω slots empty. For a set S of occupied slots, define

$$|S\rangle = \prod_{\lambda \in S} \hat{b}^\dagger_\lambda |\mathrm{vac}\rangle. \tag{B3}$$

The states $|S\rangle$ are orthonormal. The normalized permutation-symmetric state containing n pairs is

$$|D^{(\Omega)}_n\rangle = \binom{\Omega}{n}^{-1/2} \sum_{\substack{S \subseteq \{1,\ldots,\Omega\} \\ |S|=n}} |S\rangle, \qquad 0 \le n \le \Omega. \tag{B4}$$

It is the pair-slot image of the orbital state $|n\rangle$ in Eq. (2). Equivalently, $|D^{(\Omega)}_n\rangle$ is the normalized state obtained by applying $(\sum_\lambda \hat{b}^\dagger_\lambda)^n$ to $|\mathrm{vac}\rangle$; nilpotency removes repeated occupation of a slot, while the remaining terms occur with equal amplitude.

First consider the number operator $\hat{N} = \sum_\lambda \hat{n}_\lambda$, with $\hat{n}_\lambda = \hat{b}^\dagger_\lambda \hat{b}_\lambda$. Every configuration in Eq. (B4) contains exactly n occupied slots, so

$$\hat{N} |D^{(\Omega)}_n\rangle = n |D^{(\Omega)}_n\rangle. \tag{B5}$$

This proves the second identity in Eq. (23) on the symmetric subspace.

Now define the collective raising operator $\hat{B}^\dagger = \sum_\lambda \hat{b}^\dagger_\lambda$. Acting on Eq. (B4) gives

$$\hat{B}^\dagger |D^{(\Omega)}_n\rangle = \binom{\Omega}{n}^{-1/2} \sum_{|S|=n} \sum_{\lambda \notin S} |S \cup \{\lambda\}\rangle. \tag{B6}$$

Fix a final set T with $|T| = n+1$. It is produced once from each of the $n+1$ initial sets $S = T \setminus \{\lambda\}$ with $\lambda \in T$. Hence every term in the normalized state $|D^{(\Omega)}_{n+1}\rangle$ occurs $n+1$ times, and

$$\begin{aligned} \hat{B}^\dagger |D^{(\Omega)}_n\rangle &= (n+1) \sqrt{\frac{\binom{\Omega}{n+1}}{\binom{\Omega}{n}}} |D^{(\Omega)}_{n+1}\rangle \\ &= \sqrt{(\Omega - n)(n+1)}\, |D^{(\Omega)}_{n+1}\rangle. \end{aligned} \tag{B7}$$

The lowering operator $\hat{B} = \sum_\lambda \hat{b}_\lambda$ is obtained similarly. A fixed final set of $n-1$ occupied slots can be reached by removing a pair from any of the $\Omega - n + 1$ slots that were empty in that final set. Therefore

$$\begin{aligned} \hat{B} |D^{(\Omega)}_n\rangle &= (\Omega - n + 1) \sqrt{\frac{\binom{\Omega}{n-1}}{\binom{\Omega}{n}}} |D^{(\Omega)}_{n-1}\rangle \\ &= \sqrt{(\Omega - n + 1)n}\, |D^{(\Omega)}_{n-1}\rangle. \end{aligned} \tag{B8}$$

Identifying $|n\rangle$ with $|D^{(\Omega)}_n\rangle$, $\hat{P}^\dagger$ with $\hat{B}^\dagger$, $\hat{P}$ with $\hat{B}$, and $\hat{n}$ with $\hat{N}$ gives exactly Eq. (3). The same conclusion follows from the collective commutator

$$[\hat{B}, \hat{B}^\dagger] = \sum_{\lambda=1}^{\Omega} (\hat{I} - 2\hat{n}_\lambda) = \Omega \hat{I} - 2\hat{N}, \tag{B9}$$

which is the orbital quasi-spin algebra. Because the Hamiltonian contains only collective operators and treats all slots belonging to a given orbital identically, it preserves the tensor product of these orbital symmetric subspaces. Its restriction to that encoded subspace is therefore unitarily equivalent to the full seniority-zero Hamiltonian used in the main text.

A canonical boson would instead satisfy $\hat{a}^\dagger |n\rangle = \sqrt{n+1}\, |n+1\rangle$ and would allow unbounded occupation. The factor $\sqrt{\Omega - n}$ in Eq. (B7), the upper bound $n \le \Omega$, and the vanishing of the raising matrix element at $n = \Omega$ retain Pauli blocking exactly. The pair-slot representation is consequently not a bosonic approximation.

Appendix C: Direct sparse solution of the time-dependent problem

Direct sparse propagation is not a distinct many-body approximation or a standard name for one particular algorithm. In this work it describes how Eq. (13) is solved after the physical active space has been fixed: the complete coefficient vector is propagated directly, while the Hamiltonian is stored and applied as a sparse matrix. The state vector itself is generally dense. Thus "direct" does not mean analytic, and "sparse" does not imply that small amplitudes are discarded.

1. Basis and sparse Hamiltonian

Let $r = 1, \ldots, L$ run over all active target and projectile orbitals, and write a basis configuration as $\boldsymbol{n} = (n_1, \ldots, n_L)$ with $0 \le n_r \le \Omega_r$ and $\sum_r n_r = M$. We enumerate these configurations by the mixed-radix code

$$\gamma(\boldsymbol{n}) = \sum_{r=1}^{L} n_r s_r, \qquad s_1 = 1, \qquad s_r = \prod_{k=1}^{r-1} (\Omega_k + 1), \tag{C1}$$

and retain only codes with the required total pair number. The code is an indexing device: it permits the destination of an allowed pair move to be located without

searching the basis and introduces no additional physical approximation. The wave function is expanded as

$$|\Psi(x,D)\rangle = \sum_{\alpha=1}^{N_M} c_\alpha(x,D)\,|\alpha\rangle\,, \qquad x \equiv \frac{t}{\tau(D)}, \tag{C2}$$

where $N_M = 5554$ for $^{116}\mathrm{Sn} + {}^{60}\mathrm{Ni}$ and $N_M = 145$ for $^{96}\mathrm{Zr} + {}^{40}\mathrm{Ca}$.

The isolated fragment Hamiltonians are first diagonalized in their separate fixed-pair sectors. Their normalized ground states define the entrance state, which is embedded in the basis of Eq. (C2) as

$$|\Psi_{\mathrm{in}}\rangle = |\mathrm{GS}_T(M_T)\rangle \otimes |\mathrm{GS}_P(M_P)\rangle\,,$$
$$c_\alpha(-5,D) = \langle\alpha|\Psi_{\mathrm{in}}\rangle\,, \qquad \sum_{\alpha=1}^{N_M} |c_\alpha(-5,D)|^2 = 1. \tag{C3}$$

Only these time-independent fragment problems are diagonalized. During the collision the matrix representation is written as

$$\boldsymbol{H}(x,D) = \boldsymbol{H}_0 + G_{\mathrm{pk}}(D)e^{-x^2}\boldsymbol{V}_{PT},$$
$$\hat{V}_{PT} = -\sum_{\substack{p\in P\\ q\in T}} \left(\hat{P}^\dagger_{qT}\hat{P}_{pP} + \hat{P}^\dagger_{pP}\hat{P}_{qT}\right), \tag{C4}$$

where $\boldsymbol{H}_0$ contains the separated-fragment Hamiltonians and the energy alignments specified in the main text, and $\boldsymbol{V}_{PT}$ is the matrix of $\hat{V}_{PT}$. An additive entrance energy is subtracted from $\boldsymbol{H}_0$; this changes only the overall phase and improves the numerical energy scale. Equations (17)–(19) are the matrix-element formulas used in the sparse construction. A nonzero off-diagonal element connects only two configurations that differ by moving one pair between two orbitals. The matrices are assembled once in compressed sparse-row form. For $^{116}\mathrm{Sn} + {}^{60}\mathrm{Ni}$, $\boldsymbol{H}_0$ and $\boldsymbol{V}_{PT}$ contain 103174 and 110630 nonzero entries, respectively, compared with 5554^2 possible entries. The corresponding counts for $^{96}\mathrm{Zr} + {}^{40}\mathrm{Ca}$ are 1395 and 1560.

2. Adaptive time propagation

Using $x = t/\tau(D)$ converts the Schrödinger equation to

$$\frac{d\boldsymbol{c}}{dx} = -\mathrm{i}\frac{\tau(D)}{\hbar}\left[\boldsymbol{H}_0 + G_{\mathrm{pk}}(D)e^{-x^2}\boldsymbol{V}_{PT}\right]\boldsymbol{c}(x,D). \tag{C5}$$

The factor $\tau/\hbar$ is essential: τ is evaluated in seconds, $\hbar = 6.582119569 \times 10^{-22}$ MeV s, and both Hamiltonian terms are in MeV. Each evaluation of the right-hand side of Eq. (C5) therefore requires two sparse matrix–vector products and vector additions; the time-dependent matrix is neither assembled anew nor diagonalized. Time ordering is retained automatically by integrating the differential equation.

We integrate Eq. (C5) in complex double precision with the explicit eighth-order DOP853 Runge–Kutta method and adaptive error control, as implemented in SciPy [13, 14]. The reported reference trajectories and final distance-resolved observables use a relative tolerance of 3×10^{-10}, an absolute tolerance of $\max(10^{-2}\,\mathrm{rtol}, 10^{-13})$, and a maximum internal step $\Delta x = 0.05$. The transfer-curve fits use relative tolerances of 2×10^{-9} for $^{116}\mathrm{Sn}+{}^{60}\mathrm{Ni}$ and 3×10^{-9} for $^{96}\mathrm{Zr}+{}^{40}\mathrm{Ca}$, with maximum steps of 0.07 and 0.06, respectively. Requested output times determine only where the adaptive solution is recorded and do not fix the internal Runge–Kutta steps.

The integration interval is $-5 \le x \le 5$. At either endpoint, $G_{PT}/G_{\mathrm{pk}} = e^{-25} = 1.39 \times 10^{-11}$, and the fraction of the Gaussian pulse area omitted outside the interval is $\mathrm{erfc}(5) = 1.54 \times 10^{-12}$. We therefore initialize the separated product state of Eq. (C3), rather than the instantaneous ground state in the presence of the minute endpoint coupling.

3. Observables and checks

DOP853 is not algebraically unitary. We do not renormalize the state during propagation; instead, the raw norm $\mathcal{N}(x) = \boldsymbol{c}^\dagger(x)\boldsymbol{c}(x)$ is retained as an accuracy diagnostic. Let $\mathcal{I}_\kappa$ denote the basis indices for which $M_T(\alpha) = M_T^{\mathrm{in}} + \kappa$. When an observable is extracted, the transfer-sector probability is evaluated as

$$\mathcal{P}_\kappa(x,D) = \frac{1}{\mathcal{N}(x)} \sum_{\alpha\in\mathcal{I}_\kappa} |c_\alpha(x,D)|^2. \tag{C6}$$

For the entanglement calculation, the same coefficients are viewed as a fragment-product matrix $C_{\mu\nu}$, where μ and ν label target and projectile configurations. Grouping amplitudes with the same projectile index gives

$$(\rho_T)_{\mu\mu'} = \frac{1}{\mathcal{N}}\sum_\nu C_{\mu\nu}C^*_{\mu'\nu}, \qquad \mathcal{S}_{T|P} = -\sum_j \lambda_j \log_2 \lambda_j, \tag{C7}$$

where λ_j are the eigenvalues of ρ_T. This grouped construction avoids forming the mostly empty full tensor-product coefficient array and is algebraically identical to the partial trace in Eq. (22).

The tighter-tolerance comparisons quoted in Sec. III use $\mathrm{rtol} = 3 \times 10^{-12}$ and halve the maximum step. We additionally checked the sparse Hamiltonian against an independently constructed tensor-product Hamiltonian, obtaining a maximum matrix-element difference of 1.1×10^{-16} MeV, and reproduced an analytically solvable two-level Gaussian pulse to 6.7×10^{-16} in transfer probability. Finally, a separate midpoint-Magnus propagation using sparse Krylov exponential actions and step doubling agrees with the DOP853 sector distributions to within 1.8×10^{-10} for both reactions. These tests verify the implemented finite-space evolution; they do not assess omitted orbitals, broken-pair states, continuum channels, absorption, or the assumed reaction trajectory.

The term "exact evolution" used in Sec. III must therefore be read as converged propagation of every amplitude

in the declared finite seniority-zero space. Sparse storage reduces the cost of a Hamiltonian application, but N_M still grows combinatorially with the number of orbitals and pairs. Direct sparse propagation is consequently a classical benchmark for the present spaces, not evidence of quantum advantage and not an exact solution of the unrestricted nuclear reaction.

Appendix D: Poisson limit of the two-pair ratio

Consider a reference process in which pair transfers occur independently and only in the forward direction. If the mean number of transferred pairs in one collision is λ, the probability of observing exactly κ transfers is

$$\mathcal{P}^{\mathrm{Pois}}_{+\kappa} = e^{-\lambda}\frac{\lambda^{\kappa}}{\kappa!}, \qquad \kappa = 0, 1, 2, \ldots. \tag{D1}$$

This distribution follows, for example, from N independent opportunities for transfer, each with probability q, in the rare-event limit $N \to \infty$, $q \to 0$, with $Nq = \lambda$ fixed. For zero, one, and two transferred pairs, Eq. (D1) gives

$$\mathcal{P}_0 = e^{-\lambda}, \qquad \mathcal{P}_{+1} = e^{-\lambda}\lambda, \qquad \mathcal{P}_{+2} = e^{-\lambda}\frac{\lambda^2}{2}. \tag{D2}$$

The factor $1/2 = 1/2!$ is the combinatorial factor for two indistinguishable independent events. The transfer probabilities used in the main text are normalized to the elastic sector according to Eq. (21). Consequently,

$$P^{\mathrm{Pois}}_{+1} = \frac{\mathcal{P}_{+1}}{\mathcal{P}_0} = \lambda, \qquad P^{\mathrm{Pois}}_{+2} = \frac{\mathcal{P}_{+2}}{\mathcal{P}_0} = \frac{\lambda^2}{2}, \tag{D3}$$

and therefore

$$R^{\mathrm{Pois}}_2 = \frac{P^{\mathrm{Pois}}_{+2}}{(P^{\mathrm{Pois}}_{+1})^2} = \frac{\mathcal{P}_{+2}\mathcal{P}_0}{\mathcal{P}^2_{+1}} = \frac{1}{2}. \tag{D4}$$

For an exact Poisson distribution, Eq. (D4) is independent of λ. The rare-event qualification identifies the physical limit in which the Poisson model arises; it is not needed as an additional approximation after the normalization by $\mathcal{P}_0$. Without that normalization, the corresponding ratio of exclusive probabilities would be $\mathcal{P}_{+2}/\mathcal{P}^2_{+1} = e^{\lambda}/2$, which approaches $1/2$ only when $\lambda \ll 1$. Reverse transfer, a finite number of available pair slots, Pauli blocking, and coherent interference all violate the assumptions of this reference process, so the dashed line in Fig. 3 is a statistical baseline rather than a prediction of the two-fragment Hamiltonian.

[1] A. Bohr, B. R. Mottelson, and D. Pines, "Possible Analogy between the Excitation Spectra of Nuclei and Those of the Superconducting Metallic State," Phys. Rev. **110**, 936 (1958).

[2] D. M. Brink and R. A. Broglia, *Nuclear Superfluidity: Pairing in Finite Systems* (Cambridge University Press, Cambridge, 2005).

[3] D. J. Dean and M. Hjorth-Jensen, "Pairing in nuclear systems: From neutron stars to finite nuclei," Rev. Mod. Phys. **75**, 607 (2003).

[4] G. Potel, A. Idini, F. Barranco, E. Vigezzi, and R. A. Broglia, "Cooper pair transfer in nuclei," Rep. Prog. Phys. **76**, 106301 (2013).

[5] L. Corradi *et al.*, "Single and pair neutron transfers at sub-barrier energies," Phys. Rev. C **84**, 034603 (2011).

[6] D. Montanari *et al.*, "Neutron Pair Transfer in ^{60}Ni + ^{116}Sn Far below the Coulomb Barrier," Phys. Rev. Lett. **113**, 052501 (2014).

[7] R. A. Broglia and A. Winther, *Heavy Ion Reactions* (Addison-Wesley, Redwood City, 1991).

[8] E. T. Holland, K. A. Wendt, K. Kravvaris, X. Wu, W. E. Ormand, J. L. DuBois, S. Quaglioni, and F. Pederiva, "Optimal control for the quantum simulation of nuclear dynamics," Phys. Rev. A **101**, 062307 (2020).

[9] F. Turro *et al.*, "Demonstration of a quantum-classical coprocessing protocol for simulating nuclear reactions," Phys. Rev. A **108**, 032417 (2023).

[10] G. Scamps and D. Lacroix, "Effect of pairing on one- and two-nucleon transfer below the Coulomb barrier: A time-dependent microscopic description," Phys. Rev. C **87**, 014605 (2013).

[11] D. Regnier and D. Lacroix, "Microscopic description of pair transfer between two superfluid Fermi systems. II. Quantum mixing of time-dependent Hartree-Fock-Bogolyubov trajectories," Phys. Rev. C **99**, 064615 (2019).

[12] I. Abdurrahman, A. Makowski, G. Scamps, K. Godbey, and P. Magierski, "Pair Transfer and Reaction Dynamics in ^{40,48}Ca + ^{96}Zr Collisions Below the Coulomb Barrier," arXiv:2512.01808 [nucl-th] (2026).

[13] E. Hairer, S. P. Nørsett, and G. Wanner, *Solving Ordinary Differential Equations I: Nonstiff Problems*, 2nd ed. (Springer, Berlin, 1993).

[14] P. Virtanen *et al.*, "SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python," Nat. Methods **17**, 261 (2020).